



Shenzhen Hi-Link Electronic Co., Ltd.

15W DIY series AC-DC power module

15LS03/15LS05/15LS09

15LS12/15LS15/15LS24



Contents

1. Ultra-small Series Power Module	3
2. Product Model	3
3. Product features	3
4. Environmental conditions	4
5. Electrical characteristics	4
5.1. Input features	4
5.2. Output features (3.3V/3000mA)	5
5.3. Output features (5V/3000mA)	6
5.4. Output features (9V/1660mA)	7
5.5. Output features (12V/1250mA)	8
5.6. Output features (15V/1000mA)	9
5.7. Output features (24V/625mA)	10
6. Input voltage and load characteristics	11
8. Typical application circuit	11
8.1 Input parts	12
8.2 Output parts	12
8.3 Peripheral overall scheme design	13
9. Safety characteristic	14
9.1 Certification	14
9.2 Safety and electromagnetic compatibility	14
10. Marking, packaging, transportation, storage	14
10.1 Marking	14
10.1.1 Product marking	14
10.1.2 Packing marking	14
10.2 Packaging	15
10.3 Transportation	15
10.4 Storage	15
11. Dimensions and weight	16

1. Ultra-small Series Power Module

The 15W LS ultra-small series modular power supply is a compact, high-efficiency AC-DC power module developed by Shenzhen Hi-Link Electronic Co., Ltd. It features universal global input voltage range, low temperature rise, low power consumption, high efficiency, high reliability and reliable galvanic isolation, widely adopted in smart home, industrial automation control, communication equipment and instrumentation industries.

2. Product Model

MODEL	Size (mm)	Output power (W)	Output voltage (V)	Output current (mA)	Notes
HLK-15LS03	42.7*23*13.8	10	3.3	3000	
HLK-15LS05		15	5	3000	
HLK-15LS09		15	9	1660	
HLK-15LS12		15	12	1250	
HLK-15LS15		15	15	1000	
HLK-15LS24		15	24	625	

3. Product features

1. Ultra-thin, ultra-small, smallest volume in the industry;
2. Global universal input voltage (85~305Vac)
3. Low power consumption, green environmental protection, no-load loss<0.1W
4. Low ripple, low noise
5. High output short circuit and over-current protection and self recovery
6. High efficiency, high power density
7. Input and output isolation voltage 3000Vac
8. 100% full load aging and testing
9. High reliability, long life design, continuous working time is greater than 100,000 hours;
10. Meet UL, CE requirements; product design to meet EMC and safety testing requirements;
11. Economical design with high cost-effectiveness
12. 1 year quality guarantee period

4. Environmental conditions

Items	Technical Parameters	Units	Notes
Working temperature	-25—+60	°C	
Storage temperature	-40—+80	°C	
Relative humidity	5—95	%	
Cooling mode	Natural cooling		
Atmospheric pressure	80—106	Kpa	
Altitude	≤2000	m	
Vibration	Vibration coefficient 10~500Hz,2G10min./1cycle, 60min.each along X,Y,Z axes		Meets requirements for secondary road transportation

5. Electrical characteristics

5.1. Input features

Items	Technical Parameters	Units	Notes
Rated input voltage	100-240	Vac	
Input voltage range	85-305	Vac	or 100-430Vdc
The maximum input current	≤0.2	A	
Input inrush current	≤10	A	
Input low start	≤50	mS	
Long-term reliability	MTBF≥100, 000	h	
Recommended external fuse	1A / 250Vac or 10Ω wire-wound resistor		Slow-blow type

Note: Tested at room temperature

5.2. Output features (3.3V/3000mA)

Items	Technical Parameters	Units	Notes
No-load rated output voltage	3.3±0.1	Vdc	
Full-load rated output voltage	3.3±0.1	Vdc	
Short-time maximum output current	≥3850	mA	
Rated output current	3000	mA	
Voltage regulation	±0.1	%	
Load regulation	±0.5	%	
Input low voltage efficiency	Vin=115Vac, Output full load ≥82	%	
Input high voltage efficiency	Vin=230Vac, Output full load ≥81	%	
Output ripple and noise (mVp-p)	≤100 Rated input voltage, output full load. With 20MHz bandwidth oscilloscope, Load side 10uF and 0.1uF capacitance test.	mV	
Turn-on/off overshoot	(Rated input voltage, output plus 10% load) ≤ 5	%V _O	
Output over-current protection	Output maximum load 110-150%	A	
Output short circuit protection	Direct short circuit in normal output and automatic return to normal operation after removal of short circuit		No damage to the unit

5.3. Output features (5V/3000mA)

Items	Technical Parameters	Units	Notes
No-load rated output voltage	5.0±0.1	Vdc	
Full-load rated output voltage	5.0±0.1	Vdc	
Short-time maximum output current	≥3500	mA	
Rated output current	3000	mA	
Voltage regulation	±0.1	%	
Load regulation	±0.5	%	
Input low voltage efficiency	Vin=115Vac, Output full load ≥84	%	
Input high voltage efficiency	Vin=230Vac, Output full load ≥85	%	
Output ripple and noise (mVp-p)	≤100 Rated input voltage, output full load. With 20MHz bandwidth oscilloscope, Load side 10uF and 0.1uF capacitance test.	mV	
Turn-on/off overshoot	(Rated input voltage, output plus 10% load) ≤ 5	%V _O	
Output over-current protection	Output maximum load 110-150%	A	
Output short circuit protection	Direct short circuit in normal output and automatic return to normal operation after removal of short circuit		No damage to the unit

5.4. Output features (9V/1660mA)

Items	Technical Parameters	Units	Notes
No-load rated output voltage	9.0±0.2	Vdc	
Full-load rated output voltage	9.0±0.2	Vdc	
Short-time maximum output current	≥2000	mA	
Rated output current	1660	mA	
Voltage regulation	±0.2	%	
Load regulation	±0.5	%	
Input low voltage efficiency	Vin=115Vac, Output full load ≥88	%	
Input high voltage efficiency	Vin=230Vac, Output full load ≥87	%	
Output ripple and noise (mVp-p)	≤100 Rated input voltage, output full load. With 20MHz bandwidth oscilloscope, Load side 10uF and 0.1uF capacitance test.	mV	
Turn-on/off overshoot	(Rated input voltage, output plus 10% load)≤ 5	%V _O	
Output over-current protection	Output maximum load 110-150%	A	
Output short circuit protection	Direct short circuit in normal output and automatic return to normal operation after removal of short circuit		No damage to the unit

5.5. Output features (12V/1250mA)

Items	Technical Parameters	Units	Notes
No-load rated output voltage	12.0±0.2	Vdc	
Full-load rated output voltage	12.0±0.2	Vdc	
Short-time maximum output current	≥1440	mA	
Rated output current	1250	mA	
Voltage regulation	±0.2	%	
Load regulation	±0.5	%	
Input low voltage efficiency	Vin=115Vac, Output full load ≥86	%	
Input high voltage efficiency	Vin=230Vac, Output full load ≥86	%	
Output ripple and noise (mVp-p)	≤100 Rated input voltage, output full load. With 20MHz bandwidth oscilloscope, Load side 10uF and 0.1uF capacitance test.	mV	
Turn-on/off overshoot	(Rated input voltage, output plus 10% load) ≤ 5	%V _O	
Output over-current protection	Output maximum load 110-150%	A	
Output short circuit protection	Direct short circuit in normal output and automatic return to normal operation after removal of short circuit		No damage to the unit

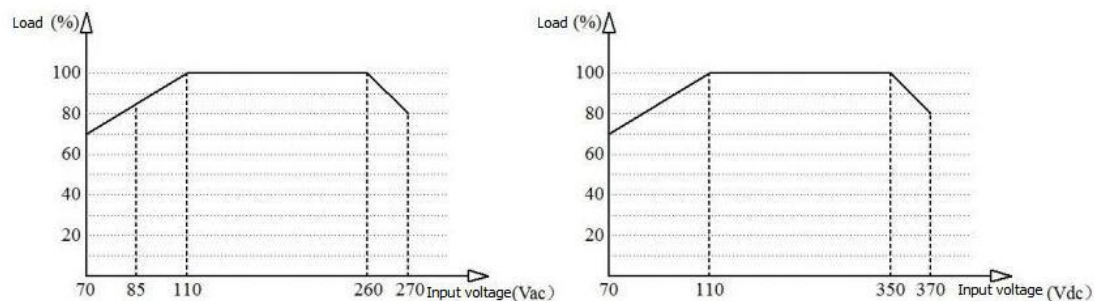
5.6. Output features (15V/1000mA)

Items	Technical Parameters	Units	Notes
No-load rated output voltage	15±0.2	Vdc	
Full-load rated output voltage	15±0.2	Vdc	
Short-time maximum output current	≥1240	mA	
Rated output current	1000	mA	
Voltage regulation	±0.2	%	
Load regulation	±0.5	%	
Input low voltage efficiency	Vin=115Vac, Output full load ≥85	%	
Input high voltage efficiency	Vin=230Vac, Output full load ≥85	%	
Output ripple and noise (mVp-p)	≤100 Rated input voltage, output full load. With 20MHz bandwidth oscilloscope, Load side 10uF and 0.1uF capacitance test.	mV	
Turn-on/off overshoot	(Rated input voltage, output plus 10% load) ≤ 5	%Vo	
Output over-current protection	Output maximum load 110-150%	A	
Output short circuit protection	Direct short circuit in normal output and automatic return to normal operation after removal of short circuit		No damage to the unit

5.7. Output features (24V/625mA)

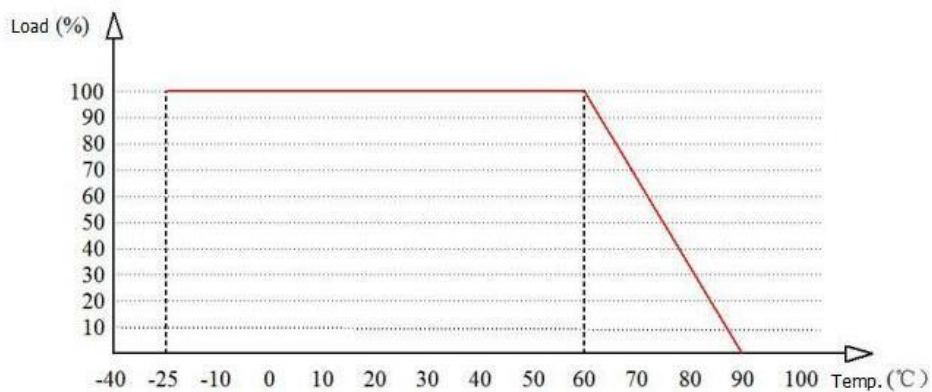
Items	Technical Parameters	Units	Notes
No-load rated output voltage	24.0±0.2	Vdc	
Full-load rated output voltage	24.0±0.2	Vdc	
Short-time maximum output current	≥750	mA	
Rated output current	625	mA	
Voltage regulation	±0.2	%	
Load regulation	±0.5	%	
Input low voltage efficiency	Vin=115Vac, Output full load ≥88	%	
Input high voltage efficiency	Vin=230Vac, Output full load ≥87	%	
Output ripple and noise (mVp-p)	≤100 Rated input voltage, output full load. With 20MHz bandwidth oscilloscope, Load side 10uF and 0.1uF capacitance test.	mV	
Turn-on/off overshoot	(Rated input voltage, output plus 10% load) ≤ 5	%V _O	
Output over-current protection	Output maximum load 110-150%	A	
Output short circuit protection	Direct short circuit in normal output and automatic return to normal operation after removal of short circuit		No damage to the unit

6. Input voltage and load characteristics

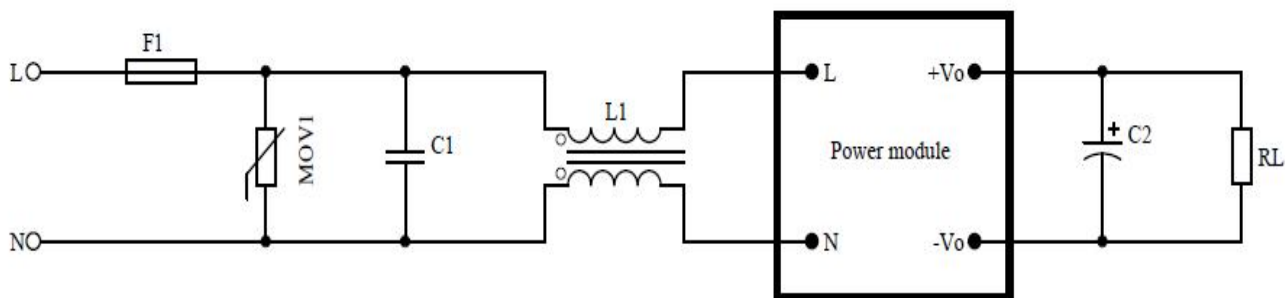


Input voltage and load characteristic curve

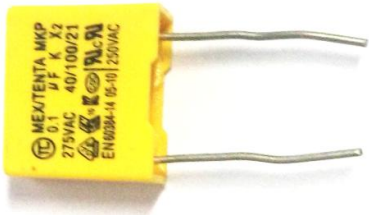
7. Derating Curve



8. Typical application circuit



8.1 Input parts

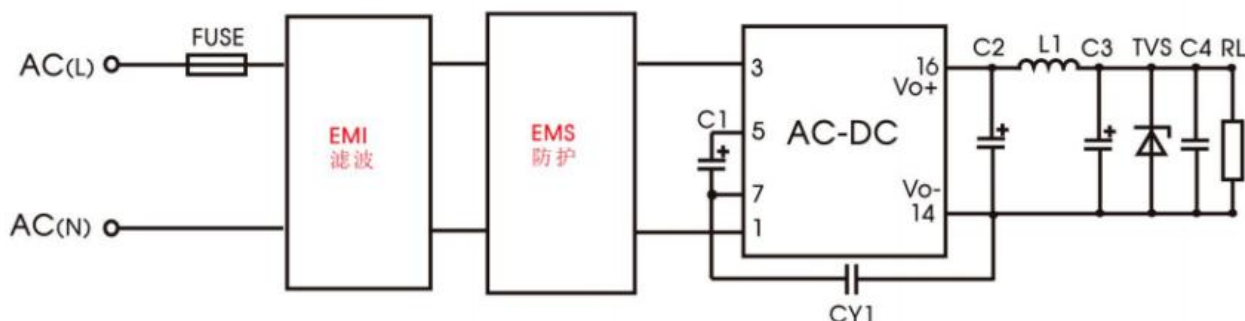
Component number / recommended device	Functions	Recommended value
F1/ Fuse	Circuit protection under abnormal module operation	1A/300Vac, Or a 10 Ω wire-wound resistor, slow-blow type
MOV1/Varistor	Surge absorption to protect module from damage	10D561K
C1/ X Safety Capacitor	Filtering, safety protection (EMC certification)	0.1uF/275Vac
L1/Common-mode inductance	EMI filtering	inductance value: 10-30mH Test requirements: 1KHZ / 0.3V current: 100-500mA
 Safety Capacitor		 Common-mode inductance

Notes:

- Fuse and varistor are mandatory basic protection components.
- Safety capacitor and common-mode choke cannot be omitted if certification is required.

8.2 Output parts

Component number / recommended device	Functions	Recommended value
C2/filter capacitor	Filter, after adding this capacitor, the user can adjust the ripple voltage of the output	Aluminum electrolytic capacitor, 100~220uF, voltage derating > 75%
RL/Load	Load	



MODEL	FUSE (must connect)	C1 (must connect)	C2 (must connect)	L1 (must connect)	C3 (must connect)	C4	CY1 (must connect)	TVS
HLK-15LS03	1A/300V	47uF/450V	1500uF/6.3V (Solid capacitor)	2.2uH (Max:8mΩ)	470uF/16V	0.1uF/ 50V	2.2nF/ 400VAC	SMBJ7.0A
HLK-15LS05			1000uF/16V (Solid capacitor)		330uF/16V			SMBJ7.0A
HLK-15LS09			680uF/16V (Solid capacitor)		220uF/35V			SMBJ12A
HLK-15LS12			1000uF/35V					SMBJ20A
HLK-15LS15			470uF/35V					SMBJ20A
HLK-15LS24								SMBJ30A

9.Safety characteristic

9.1 Certification

Product design meets UL and CE safety certification requirements. (The UL and CE certifications are made by the customer and need to be designed according to the reference circuit.)

9.2 Safety and electromagnetic compatibility

- The input is designed with UL listed 1A/300Vac slow-blow fuses or 10Ω wire-wound resistors.
- The PCB board is made of double-sided copper clad foil, and the material fire resistance grade is 94-V0 grade
- Safety standard meets UL1012,EN60950,UL60950
- Insulation voltage I/P-O/P:2500Vac
- Insulation resistance I/P-O/P>100M Ohms/500Vdc 25°C 70% RH
- Conduction and radiation meet EN55011, EN55022 (CISPR22)
- Electrostatic discharge IEC/EN 61000-4-2 level 4 8kV/15kV
- Radio frequency radiation immunity IEC/EN 61000-4-3

10.Marking, packaging, transportation, storage

10.1 Marking

10.1.1 Product marking

The product's unique bar code mark is attached to the appropriate location of the product to ensure traceability of the date of manufacture, product batch, etc. of each product. Its content meets the requirements of national standards and industry standards.

10.1.2 Packing marking

Product box marked with the name of the manufacturer, site, zip code, product model, factory year, month, day; Marked with "This Side Up", "moisture-proof" and "Handle with care" and other transport signs, all signs are in line with the provisions of GB 191.

10.2 Packaging

Individually packed with customized anti-vibration plastic tray, with anti-vibration function, and in line with the provisions of GB 3873.

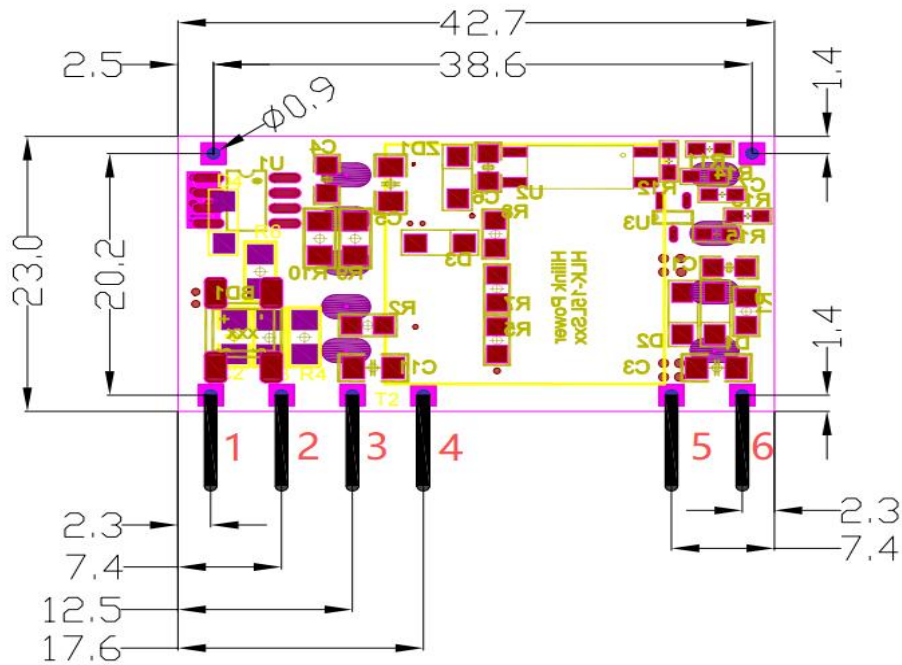
10.3 Transportation

Packaged products can be transported by any means of transportation, covered with tarpaulin during transportation, there should be no violent vibration, impact, etc.

10.4 Storage

Product storage must meet the requirements of GB3873.

11. Dimensions and weight



Dimensional error:

1. Length, width, height and pin pitch error $\pm 0.5\text{mm}$
2. Pin length error $\pm 0.2\text{mm}$

Pin function	
1	AC
2	AC
3	+V(cap)
4	-V(cap)
5	-V0
6	+V0
Weight: 12.0 $\pm$ 0.2g	